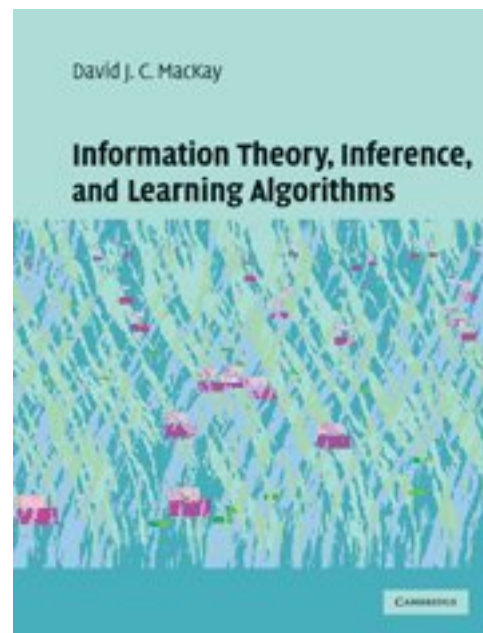
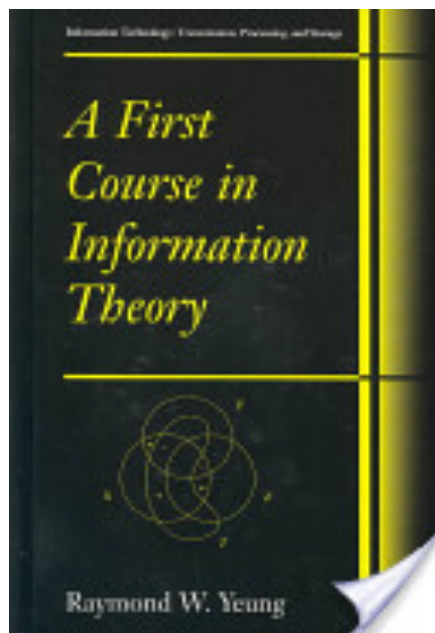
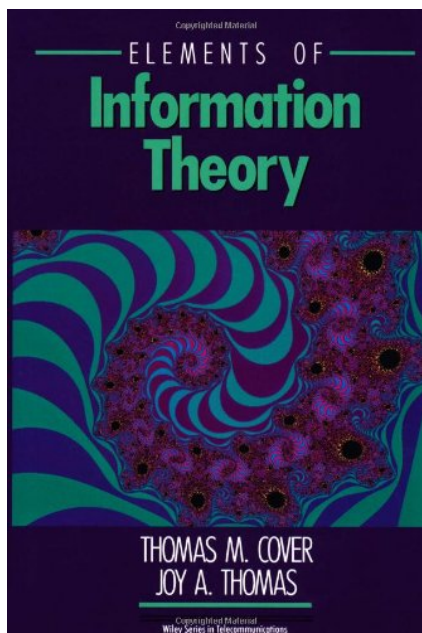

Data Science

S. Voloshynovskiy

Information Theory in data science

Recommended books



Course Outline

- Entropy, conditional entropy, joint entropy
- KLD
- Cross-entropy
- Mutual information
- Particularities of IT measures for continuous random variables

Recall of probability

Chain rule for probability: joint probability $p(x_1, x_2, \dots, x_n)$

$$\begin{aligned} p(x_1, x_2, \dots, x_n) &= p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_n | x_{n-1}, \dots, x_2, x_1) = \\ &= \prod_{i=1}^n p(x_i | x_{i-1}, \dots, x_2, x_1) \end{aligned}$$

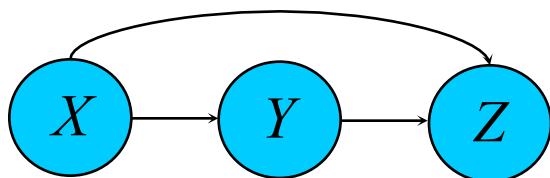
$$p(x_1, x_2) = p(x_1) p(x_2 | x_1) \xrightarrow{\text{independence}} p(x_1, x_2) = p(x_1) p(x_2)$$

$$p(x_1, x_2, \dots, x_n) = p(x_1) p(x_2) p(x_3) \dots p(x_n) = \prod_{i=1}^n p(x_i)$$

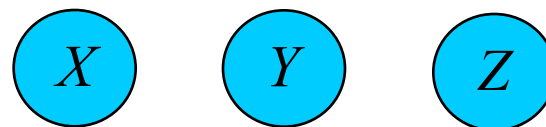
For the independent R.V.s $\{X_i\}$

$$p(x_1, x_2, \dots, x_n) = p(x) p(x) p(x) \dots p(x) = \prod_{i=1}^n p(x) = (p(x))^n$$

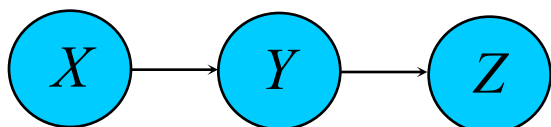
Summary on chain rule



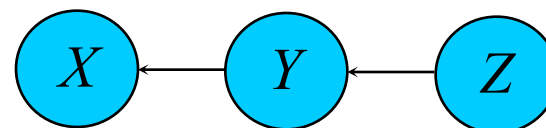
$$p(x, y, z) = p(x)p(y|x)p(z|x, y)$$



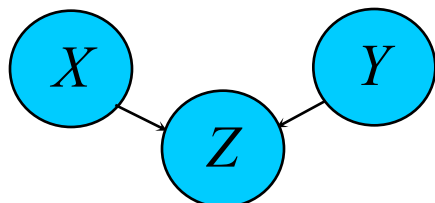
$$p(x, y, z) = p(x)p(y)p(z)$$



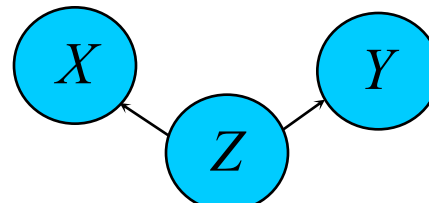
$$p(x, y, z) = p(x)p(y|x)p(z|y)$$



$$p(x, y, z) = p(z)p(y|z)p(x|y)$$



$$p(x, y, z) = p(z|x, y)p(x)p(y)$$



$$p(x, y, z) = p(z)p(x|z)p(y|z)$$

Entropy

Definition (entropy): *Entropy* of discrete r.v. $X \in \mathcal{X}$ with $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$

$$H(X) = -\sum_{x \in \mathcal{X}} p_X(x) \log_2 p_X(x) = -E_{p_X} [\log p_X(x)] \quad (\text{bits})$$

$$H(X) = -\sum_{i=1}^n \Pr\{X = x_i\} \log_2 \Pr\{X = x_i\} \quad E_{p_X} [g(X)] = \sum_{x \in \mathcal{X}} p_X(x) g(x)$$

- When the base of logarithm is α , we use the notation $H_\alpha(X)$
- Entropy measures the **uncertainty** or **randomness** of a discrete random variable
- The units of entropy
 - bit – if $\alpha = 2$
 - nat – if $\alpha = e$
- $H(X)$ depends only on the distribution of X but **not** on the actual value taken by X , hence also it is reflected as $H(p)$ or $H(p(x))$

Entropie: **binary** random variable

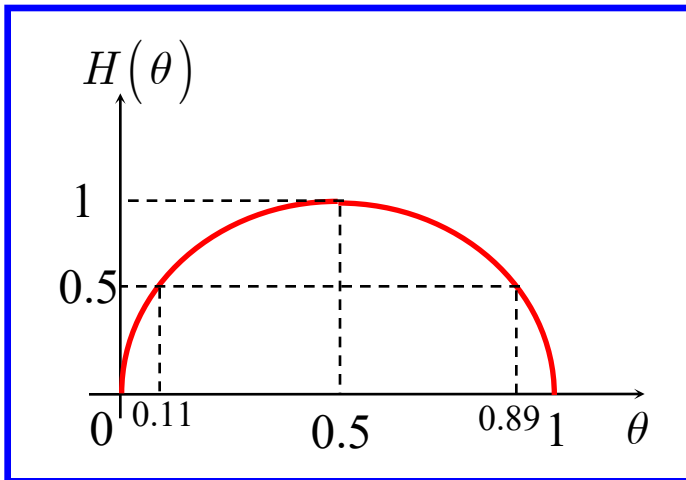
- **Binary random variable** $X \in \mathcal{X} = \{0,1\}$

$$\Pr\{X = 0\} = 1 - \theta$$

$$\Pr\{X = 1\} = \theta$$

$$\Rightarrow \Pr[X = x] = \theta^x (1 - \theta)^{1-x}$$

$$H(X) = -\theta \log_2 \theta - (1 - \theta) \log_2 (1 - \theta) := H(\theta) \text{ or } H_2(\theta)$$

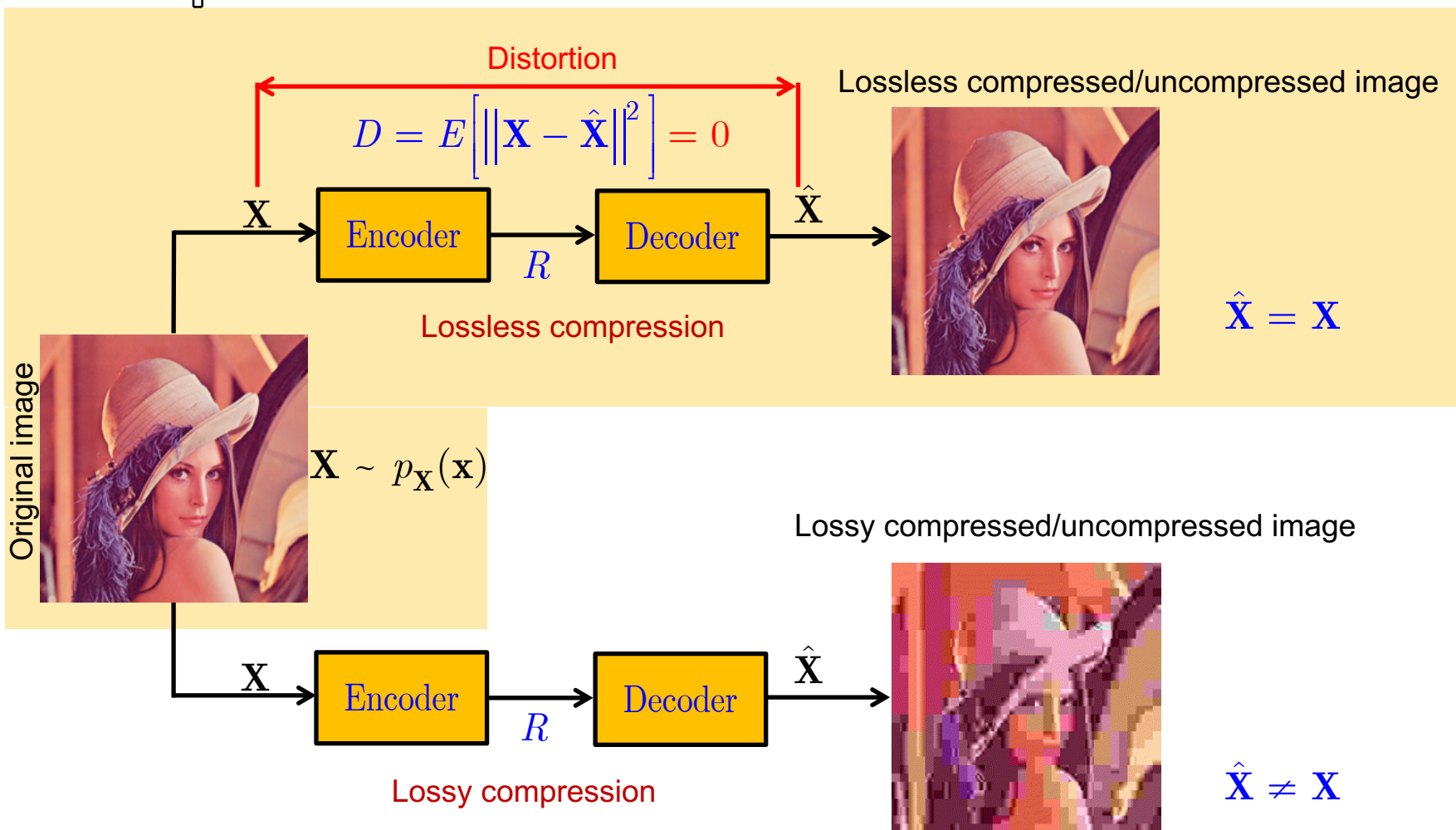


The function $H(\theta)$ is:

- Symmetric wrt $p = 0.5$
- Maximum ($H(p) = 1$) at $\theta = 0.5$

The entropy is maximal, if the symbols are equiprobable.

Entropy: Application for **lossless** compression



Interpretation of image “complexity” for coding

Summary

Knowledge of source pdf/pmf

- ① The knowledge of joint pdf $\mathbf{x} \sim p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{x}}(x_1, x_2, \dots, x_n)$ is needed for the compression
- ② Modeling issue: the multidimensional pdf is very complex
- ③ The true pdf is unknown
- ④ A common simplification: consider pixels/symbols independently – however it increases the entropy
- ⑤ Independent coding of symbols is not optimal in view of Shannon rate-distortion theory that suggests a block-based coding
- ⑥ Intermediate solution: use sub-block coding (like in JPEG as we will see it)

Complexity

- ① Shannon block coding – exponential complexity in terms of block length
- ② Practical methods – reduce the block size up to one symbol/pixel (suboptimal but lower complexity)

Compression ratio

Practical approach to measure the performance, one uses the compression ratio:

$$C_R = \frac{\text{Number of bits before}}{\text{Number of bits after}}$$

The compression ratio for real images:

- Lossless – about 2
- Lossy – 10-50 (JPEG) and up to 150 (JPEG 2000)

Compression link to IT: optimal code assignment

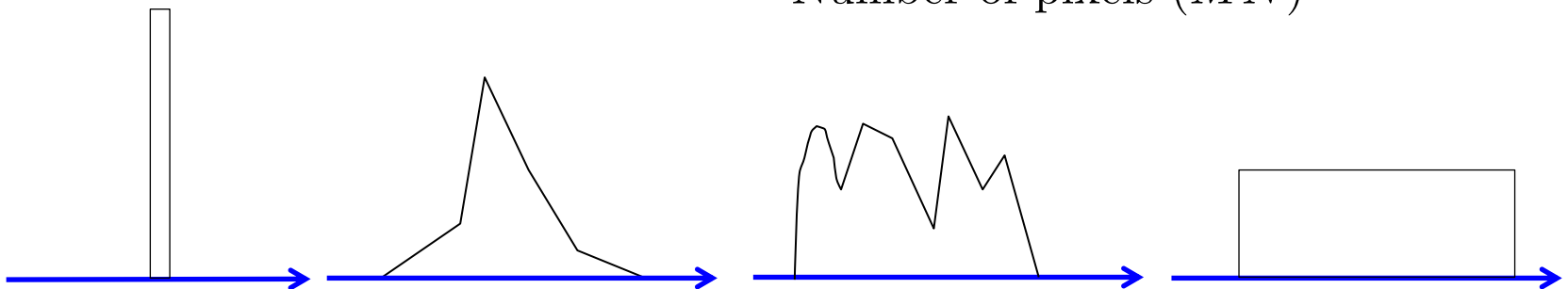
- Histogram is very misleading but let assume the image pixels are discrete and independent.
- The entropy for the discrete random variables is defined as:

$$H(X) = - \sum_{i=0}^{L-1} p_i \log p_i \triangleq \ell_{av}$$

This corresponds to **the average number of bits** needed to describe each pixel.

- The probability of observing a particular intensity in the image:

$$p_i = \Pr\{X(i, j) = k\} = \frac{\text{Number } x(i, j) = k}{\text{Number of pixels } (MN)}$$



Compression link to IT: optimal code assignment

Link between the probability and the length of the code

Define a length of code for each gray-scale level with the probability p_i as:

$$\ell_i \triangleq -\log p_i = \log \frac{1}{p_i}$$

$$\ell_{av} = H(X) = - \sum_{i=0}^{L-1} p_i \log p_i = \underbrace{\sum_{i=0}^{L-1} p_i \ell_i}_{\text{expected/average length}}$$

$$p_i = 0.5 = 2^{-1} : \ell_i = \log \frac{1}{p_i} = \log \frac{1}{2^{-1}} = \log 2 = 1$$

$$p_i = 0.25 = 2^{-2} : \ell_i = \log \frac{1}{p_i} = \log \frac{1}{2^{-2}} = \log 2^2 = 2$$

Conclusion:

More frequent symbols are assigned shorter codes, less frequent symbols are assigned longer codes == variable length coding

Compression link to IT: optimal code assignment

Consider an 8-level image (L=8)

Level	Probability	Fixed length code	Variable length code
0	0.19	000	00
1	0.25	001	11
2	0.21	010	01
3	0.16	011	101
4	0.08	100	1001
5	0.06	101	10001
6	0.03	110	100001
7	0.02	111	100000

Compute the entropy of source and the average length ℓ_{av} of these codes

Compression link to IT: optimal code assignment

```
1 % Entropy and average rate for the fixed- and varying-length codes
2
3 - p=[0.19 0.25 0.21 0.16 0.08 0.06 0.03 0.02];
4 - Sum_prob=sum(p)
5
6 % Entropy
7 - H = - sum( p.*log2(p) )
8
9 % Fixed length
10 - l_av_fixed = sum( 3*p)
11
12 % Var length
13 - Length_var= [2 2 2 3 4 5 6 6];
14 - l_av_var = sum( Length_var.* p)
15
```

Sum_prob =

1

H =

2.6508

l_av_fixed =

3

l_av_var =

2.7000

>>

- ① The assigned varying-length code is very close to the source entropy
- ② The fixed-length code is far away from the entropy
- ③ The difference might be significant for large file sizes:
 - $3 - 2.65 = 0.35$ bits per symbol/pixel
 - Assume $1000 \times 1000 = 1\text{Mio}$ symbols
 - The difference is $1\text{Mio} \times 0.35 = 3.5 \times 10^5$ bits = 42.7 Kbytes

Practical source codes approaching entropy

How to design code that approaches the entropy of source?

Our assumptions:

- We again assume that all symbols are independent (not very realistic)
- We will encode symbol by symbol
- We assume that the source pmf is known (not very realistic but can be estimated as a histogram)

Huffman coding: optimal (minimum code length) for independent coding N symbols

Huffman coding

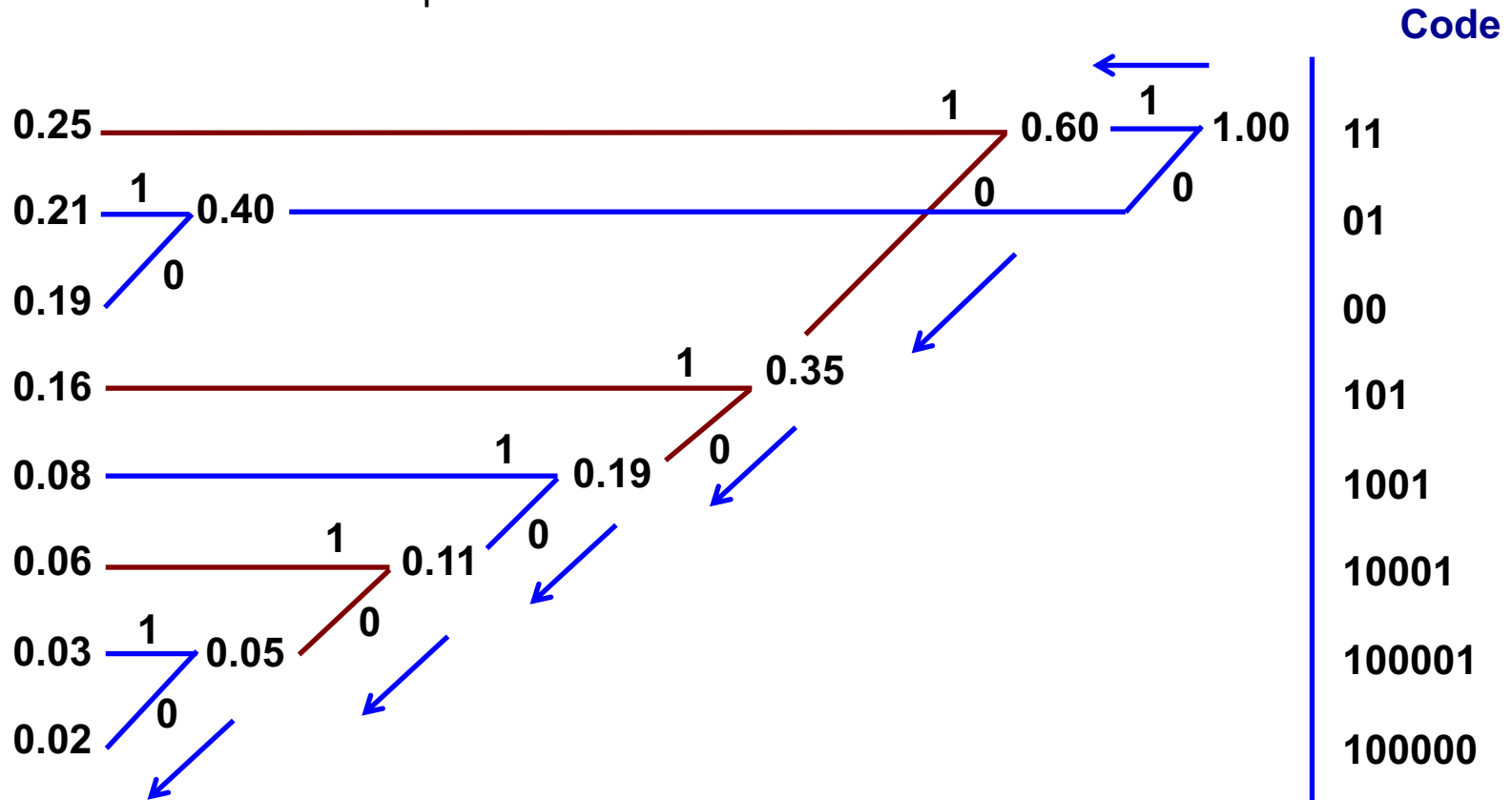
Huffman coding algorithm:

- Arrange symbols in decreasing p_i . Consider them as nodes
- Merge the nodes with the lowest probabilities
- Assign 0 and 1 to the top and bottom branches, accordingly
- Read the code from the root to leaf

Level	Probability	Fixed length code	Variable length code
0	0.19	000	00
1	0.25	001	11
2	0.21	010	01
3	0.16	011	101
4	0.08	100	1001
5	0.06	101	10001
6	0.03	110	100001
7	0.02	111	100000

General image transformation

- Consider the ordered probabilities



The symbols with the highest probability have the shortest length.

Huffman code

Huffman code is a prefix code: no codeword is a prefix of any other code.

Why is so important?

1. For the unique decoding from a string without a need to see a next possible symbol!
2. The prefix code can be decoded instantaneously.

Code A (not prefix code)

1	10
2	00
3	11
4	110

Code B (prefix code)

1	0
2	10
3	110
4	111

Example: 1234321

A: 100011110110010

B: 010110111110100

- No unique decoding

- Unique decoding; just see the next symbol and decide

Joint entropy

Definition (joint entropy): *Joint entropy* of two discrete random variables X and Y is defined by:

$$H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X,Y}(x, y) \log_2 p_{X,Y}(x, y) = -E_{p_{X,Y}} [\log p_{X,Y}(x, y)]$$

$$\mathcal{X} = \{x_1, x_2, \dots, x_n\} \quad \mathcal{Y} = \{y_1, y_2, \dots, y_m\}$$

$$H(X, Y) = H(Y, X)$$

Conditional entropy

Definition (conditional entropy): *Conditional entropy* of r.v. X given Y is defined by:

$$H(X|Y) = \sum_{y \in \mathcal{Y}} p_Y(y) H(X|Y=y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X,Y}(x,y) \log p_{X|Y}(x|y)$$

$$\mathcal{X} = \{x_1, x_2, \dots, x_n\} \quad \mathcal{Y} = \{y_1, y_2, \dots, y_m\}$$

$$H(X|Y) \neq H(Y|X)$$

$$H(X|Y=y) = - \sum_{i=1}^n p_{X|Y}(x_i|y) \log p_{X|Y}(x_i|y)$$

$$H(X|Y) = - \sum_{j=1}^m p_Y(y_j) \left[\sum_{i=1}^n p_{X|Y}(x_i|y_j) \log p_{X|Y}(x_i|y_j) \right] =$$

$$= - \sum_{i=1}^n \sum_{j=1}^m p_{X,Y}(x_i, y_j) \log p_{X|Y}(x_i|y_j)$$

Summary: properties of entropy

- **Property 1: non-negativity of entropy**
- **Property 2: chain rule for entropy**
- **Property 3: increase of entropy by “adding” new randomness**
- **Property 4: decrease of entropy by conditioning**
- **Property 5: uniform pmf has the largest entropy**

Summary: properties of entropy

- Property 1: entropy is strictly positive

$$H(X) \geq 0$$

$$0 \leq p_X(x) \leq 1 \quad \log p_X(x) \leq 0$$

$$H(X, Y) \geq 0$$

$$0 \leq p_{X,Y}(x, y) \leq 1 \quad \Rightarrow \quad \log p_{X,Y}(x, y) \leq 0$$

$$H(X|Y) \geq 0$$

$$0 \leq p_{X|Y}(x|y) \leq 1 \quad \log p_{X|Y}(x|y) \leq 0$$

$$0 \leq -\sum p_X(x) \log p_X(x)$$

$$\Rightarrow 0 \leq -\sum p_{X,Y}(x, y) \log p_{X,Y}(x, y)$$

$$0 \leq -\sum p_{X|Y}(x|y) \log p_{X|Y}(x|y)$$

Summary: properties of entropy

- **Property 2: chain rule for entropy (case of 2 variables)**

$$p(x, y) = p(x|y)p(y) = p(y|x)p(x) \Leftrightarrow p_{X,Y}(x, y) = p_{X|Y}(x|y)p_Y(y) = p_{Y|X}(y|x)p_X(x)$$

$$H(X, Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$$

Proof

$$-\log p_{X,Y}(x, y) = -\log p_{Y|X}(y|x)p_X(x) = -\log p_{Y|X}(y|x) - \log p_X(x).$$

$$\begin{aligned} H(X, Y) &= E_{p_{X,Y}}[-\log p_{X,Y}(x, y)] = E_{p_{X,Y}}[-\log p_X(x)] + E_{p_{X,Y}}[-\log p_{Y|X}(y|x)] \\ &= H(X) + H(Y|X). \end{aligned}$$

Summary: properties of entropy

- Property 2: chain rule for entropy (case of N variables)

$$p(x_1, \dots, x_N) = p(x_1) \prod_{i=2}^N p(x_i | x_{i-1}, \dots, x_1)$$

$$H(X_1, X_2, \dots, X_N) = \sum_{i=1}^N H(X_i | X_{i-1}, \dots, X_1)$$

Proof

$$H(X_1, X_2) = H(X_1) + H(X_2 | X_1) \quad \Rightarrow \quad H(X_1, X_2) \leq H(X_1) + H(X_2) \quad \downarrow \text{=, if independent}$$

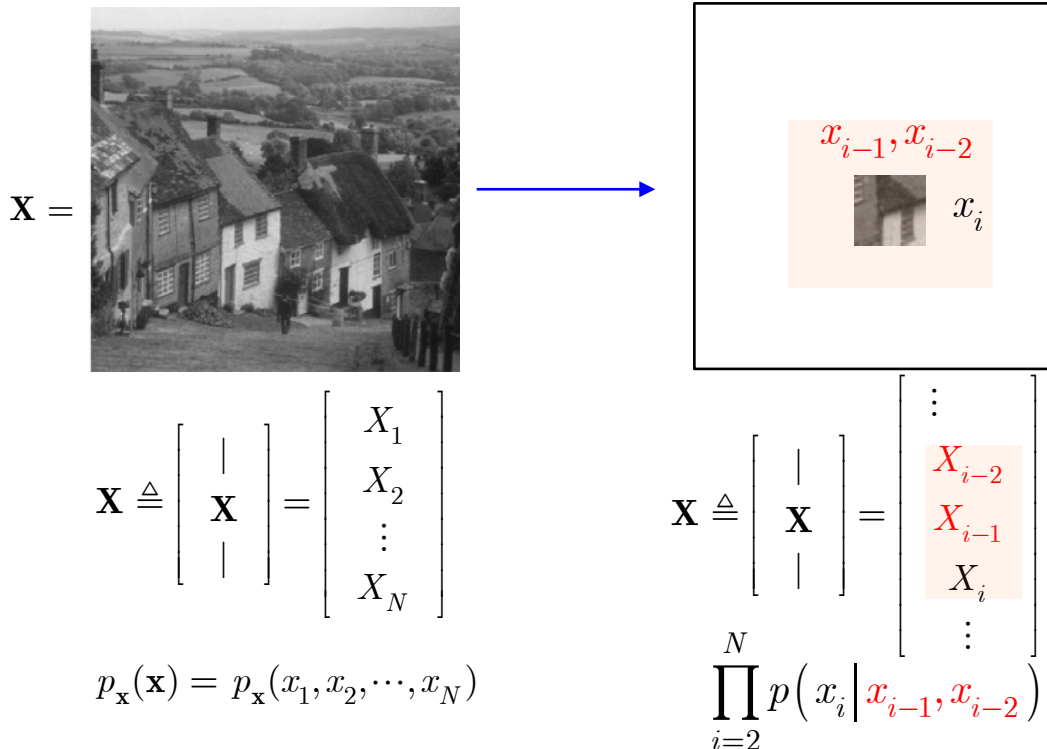
$$H(X_1, X_2, X_3) = H(X_1) + H(X_2, X_3 | X_1) = H(X_1) + H(X_2 | X_1) + H(X_3 | X_2, X_1)$$

$$\begin{aligned} H(X_1, X_2, \dots, X_N) &= H(X_1) + H(X_2 | X_1) + \dots + H(X_N | X_{N-1}, \dots, X_1) = \\ &= \sum_{i=1}^N H(X_i | X_{i-1}, \dots, X_1) \end{aligned}$$

Summary: properties of entropy

- Property 2: chain rule for entropy (case of N variables)

$$H(X_1, X_2, \dots, X_N) = \sum_{i=1}^N H(X_i | \underbrace{X_{i-1}, \dots, X_1}_{\text{past}})$$



Summary: properties of entropy

- Property 2: chain rule for entropy (case of N variables)

$$H(X_1, X_2, \dots, X_N) \leq \sum_{i=1}^N H(X_i)$$

equality for independent $\{X_i\}$

Proof

For Independent $\{X_i\}$

$$H(X_2 | X_1) = H(X_2)$$

$$H(X_3 | X_2, X_1) = H(X_3)$$

$$H(X_i | X_{i-1}, \dots, X_1) = H(X_i)$$

$$H(X_1, X_2, \dots, X_N) = H(X_1) + H(X_2 | X_1) + \dots + H(X_N | X_{N-1}, \dots, X_1) =$$

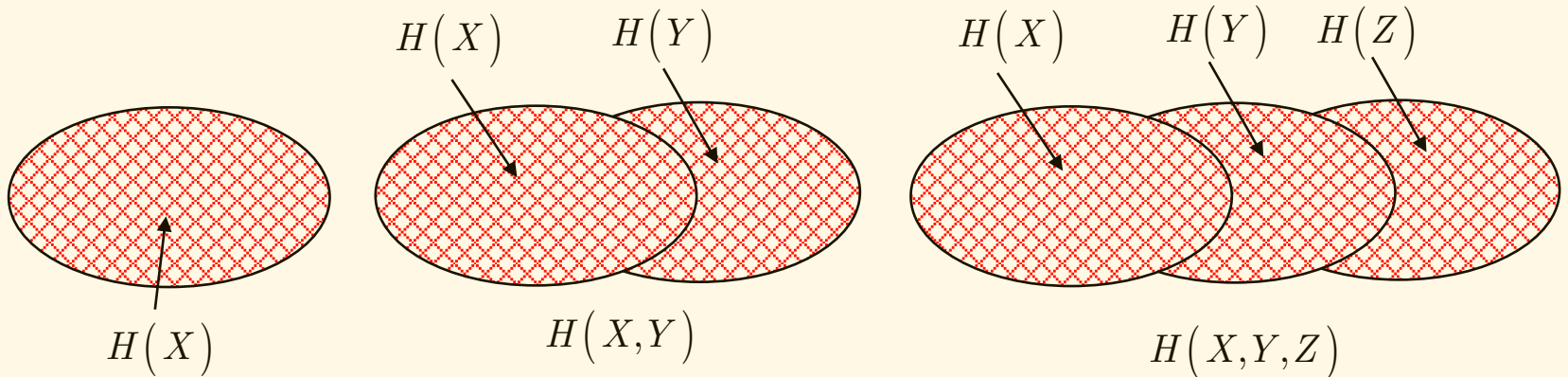
$$H(X_1, X_2, \dots, X_N) = H(X_1) + H(X_2) + \dots + H(X_N) = \sum_{i=1}^N H(X_i)$$

Summary: properties of entropy

- Property 3: increase of entropy by adding new randomness

$$H(X) \leq H(X,Y) \leq H(X,Y,Z) \leq \dots$$

$$H(X) = H(X,Y) \Leftrightarrow Y = f(X) \quad H(X,Y) = H(X) + \underbrace{H(Y|X)}_0$$
$$H(X,Y) = H(X,Y,Z) \Leftrightarrow Z = f(X,Y)$$



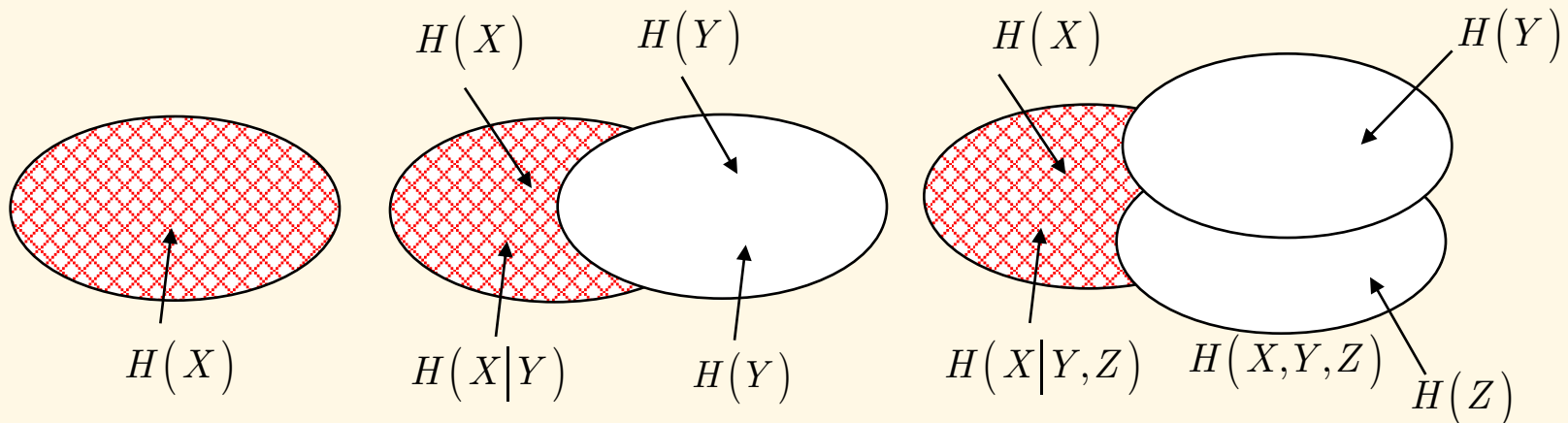
Summary: properties of entropy

- Property 4: conditioning reduces entropy

$$H(X) \geq H(X|Y) \geq H(X|Y, Z) \geq \dots$$

$$H(X) = H(X|Y) \Leftrightarrow X \perp Y$$

$$H(X|Y) = H(X|Y, Z) \Leftrightarrow X \perp Z|Y \dots$$



Summary: properties of entropy

- **Property 5: uniform pmf has the largest entropy**

$$H(X) \leq \log_2 |\mathcal{X}|$$

equality, iff $\Pr[X = x_i] = \frac{1}{N} = \frac{1}{|\mathcal{X}|}$

- **Proof**

Let $\mathcal{X} = \{x_1, x_2, \dots, x_N\}$ and $\Pr[X = x_i] = p_i$

$$\begin{aligned} H(X) &= -\sum_{x \in \mathcal{X}} \Pr[X = x] \log_2 \Pr[X = x] = \\ &= -(p_1 \log_2 p_1 + p_2 \log_2 p_2 + \dots + p_N \log_2 p_N) \end{aligned}$$

- **Our goal:** to find a distribution $\{p_i\}_{i=1}^N$ that has the maximum entropy $H(X)$

Relative entropy

Definition (relative entropy): *Relative entropy* or *Kullback-Leibler divergence (KLD)* between pmfs $p(x)$ and $q(x)$:

$$D_{\text{KL}}(p \parallel q) = D(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{p(x)}{q(x)} = E_{p(x)} \left[\log_2 \frac{p(x)}{q(x)} \right]$$

with the conventions: $0 \log_2 \frac{0}{q(x)} = 0, \forall q(x)$

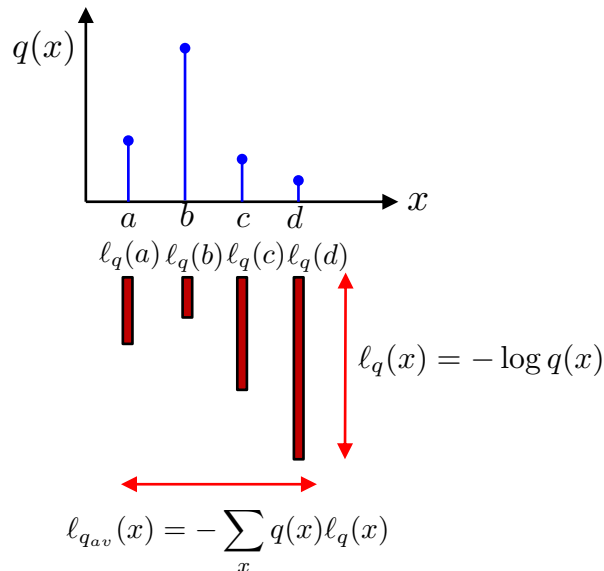
$$p(x) \log_2 \frac{p(x)}{0} = +\infty, \forall p(x) > 0.$$

$$D(p \parallel q) \neq D(q \parallel p)$$

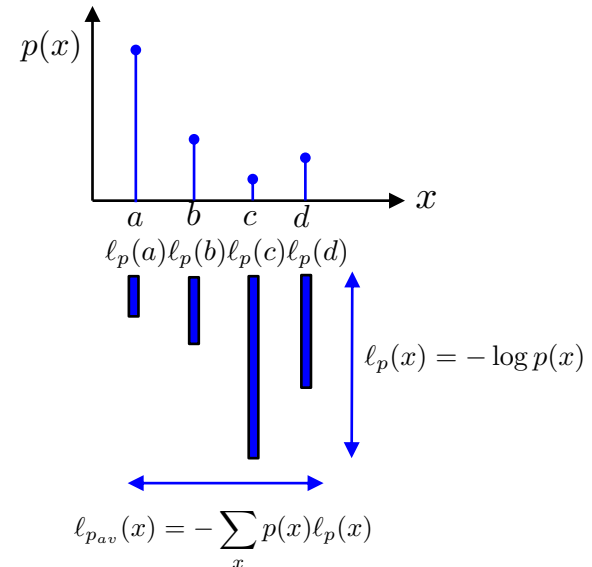
$$D(p \parallel q) \geq 0$$

Relative entropy

Assumed data distribution $q(x)$



Data distribution $p(x)$

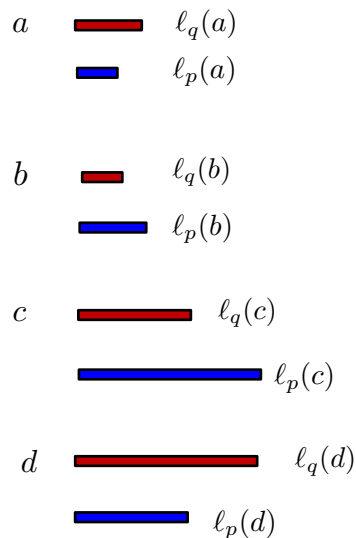


Practical problem (the true distribution $p(x)$ is unknown)

Design code for $q(x)$ $\longrightarrow$ Use the designed code for $q(x)$ to code symbols from $p(x)$

Relative entropy

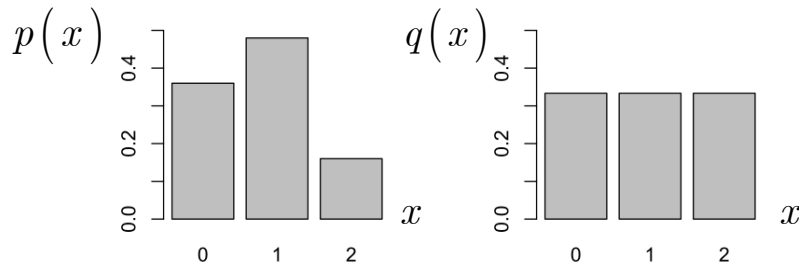
Under this assumption each symbol will be coded “suboptimally”



Question: what is the “overall loss” in this mismatched usage of codes for different sources?

Relative entropy: interpretation

- Given two distributions



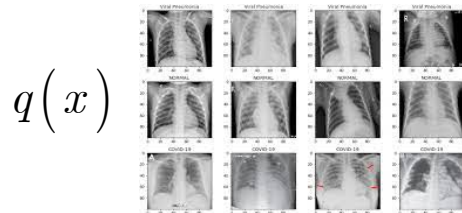
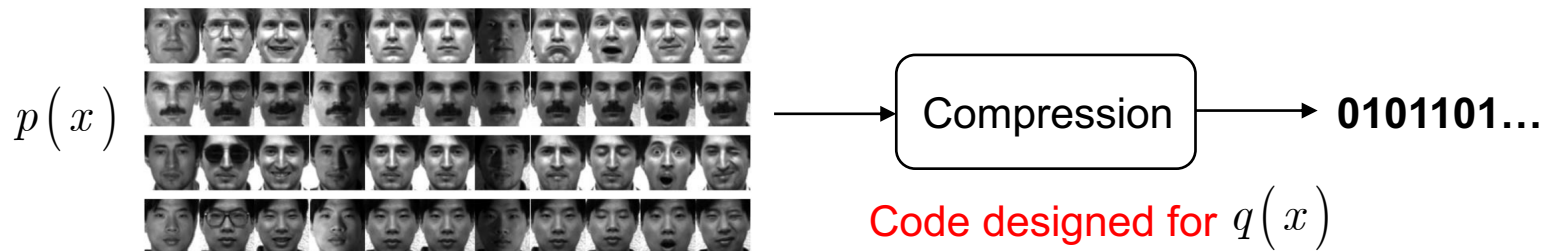
- KLD “measures” in:
 - Machine learning**: the level of “similarity” between these two distributions
 - Coding theory**: “inefficiency” of coding designed for the pmf $q(x)$ while actually data originates from pmf $p(x)$

Real data $X \sim p(x)$

Code design $H(p(x)) + D(p(x) \parallel q(x))$
extra bits

$$D(p(x) \parallel q(x)) = 0 \text{ iff } p(x) = q(x)$$

Relative entropy: interpretation



Real data	$X \sim p(x)$
Code design	$H(p(x)) + \underbrace{D(p(x) \parallel q(x))}_{\text{extra bits}}$

Relative entropy: interpretation

- Allow fractional code lengths $\ell_q(x) = -\log q(x)$
- Then expected length for coding $X \sim p(x)$ using $\ell_q(x)$ is

$$\begin{aligned} L &= \mathbb{E}_{X \sim p(x)}[\ell_q(X)] \\ &= - \sum_x p(x) \log q(x) \quad \longrightarrow \quad \text{See later as cross-entropy} \\ &= \sum_x p(x) \log \left[\frac{p(x)}{q(x)} \frac{1}{p(x)} \right] \\ &= \sum_x p(x) \log \frac{p(x)}{q(x)} + \sum_x p(x) \log \frac{1}{p(x)} \\ &= D(p(x) \| q(x)) + H(p(x)) \end{aligned}$$

Relative entropy

Example:

$$x \in \mathcal{X} = \{0,1\}; p(0) = a; p(1) = 1 - a; q(0) = b; q(1) = 1 - b.$$

$$D(p\|q) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{p(x)}{q(x)} = a \log_2 \frac{a}{b} + (1 - a) \log_2 \frac{1 - a}{1 - b};$$

$$D(q\|p) = \sum_{x \in \mathcal{X}} q(x) \log_2 \frac{q(x)}{p(x)} = b \log_2 \frac{b}{a} + (1 - b) \log_2 \frac{1 - b}{1 - a}.$$

$$a = \frac{1}{4}; b = \frac{1}{8}, D(p\|q) = \frac{1}{4} \log_2 \frac{8}{4} + \left(1 - \frac{1}{4}\right) \log_2 \left(\frac{1 - \frac{1}{4}}{1 - \frac{1}{8}} \right) = 0.0832 \text{ bit};$$

$$D(p\|q) \neq D(q\|p)$$

$$D(q\|p) = \frac{1}{8} \log_2 \frac{4}{8} + \left(1 - \frac{1}{8}\right) \log_2 \left(\frac{1 - \frac{1}{8}}{1 - \frac{1}{4}} \right) = 0.0696 \text{ bit}.$$

Cross-entropy

Definition (cross-entropy): *Cross-entropy* between pmfs $p(x)$ and $q(x)$

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = -E_{p(x)}[\log q(x)]$$

- A link to KLD:

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$$

- Proof:

$$\begin{aligned} D_{\text{KL}}(p \parallel q) &= E_{p(x)} \left[\log \frac{p(x)}{q(x)} \right] = E_{p(x)} [\log p(x)] - E_{p(x)} [\log q(x)] \\ &= -H(p) + H(p; q) \end{aligned}$$

- Remark: since $H(p)$ is constant, optimizing $H(p, q)$ wrt $q(x)$ is equivalent to $D_{\text{KL}}(p \parallel q)$

Mutual information

Definition (mutual information): *Mutual information* between two r.v. X and Y is defined by:

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X,Y}(x,y) \log \frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} = E_{p_{X,Y}(x,y)} \left[\log \frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} \right]$$

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X,Y}(x,y) \log \frac{p_{X|Y}(x|y) \cancel{p_Y(y)}}{p_X(x) \cancel{p_Y(y)}}$$

$$I(X;Y) = I(Y;X)$$

$$I(X;Y) \geq 0$$

$$\Rightarrow I(X;Y) = 0 \Leftrightarrow X \perp Y \quad (\text{if independent})$$

Mutual information

Mutual information via KL-divergence

$$I(X;Y) = D_{\text{KL}}(p_{X,Y}(x,y) \| p_X(x)p_Y(y)) = E_{p_{X,Y}(x,y)} \left[\log \frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} \right]$$

$$I(X;Y) = \mathbb{E}_{p(y)} [D_{\text{KL}}(p(x|y) \| p(x))] = \mathbb{E}_{p(x)} [D_{\text{KL}}(p(y|x) \| p(y))]$$

- Mutual information measures how different the joint pmf is from the product of marginal pmfs, i.e., a **real pmf**, vs an assumption of **independent variables**

Pay attention: on the equivalence of notations $p_{X,Y}(x,y) = p(x,y)$

Mutual information

Mutual information via KL-divergence

$$\begin{aligned} I(X; Y) &= D_{\text{KL}}(p(x, y) \| p(x)p(y)) = \mathbb{E}_{p(x, y)} \left[\log \frac{p(x, y)}{p(x)p(y)} \right] \\ &= \mathbb{E}_{p(x)p(y|x)} \left[\log \frac{p(y | x)p(x)}{p(x)p(y)} \right] = \mathbb{E}_{p(x)} \left[\mathbb{E}_{p(y|x)} \left[\log \frac{p(y | x)}{p(y)} \right] \right] \\ &= \mathbb{E}_{p(x)} [D_{\text{KL}}(p(y | x) \| p(y))] \end{aligned}$$

Mutual Information

Relationship between mutual information and entropy

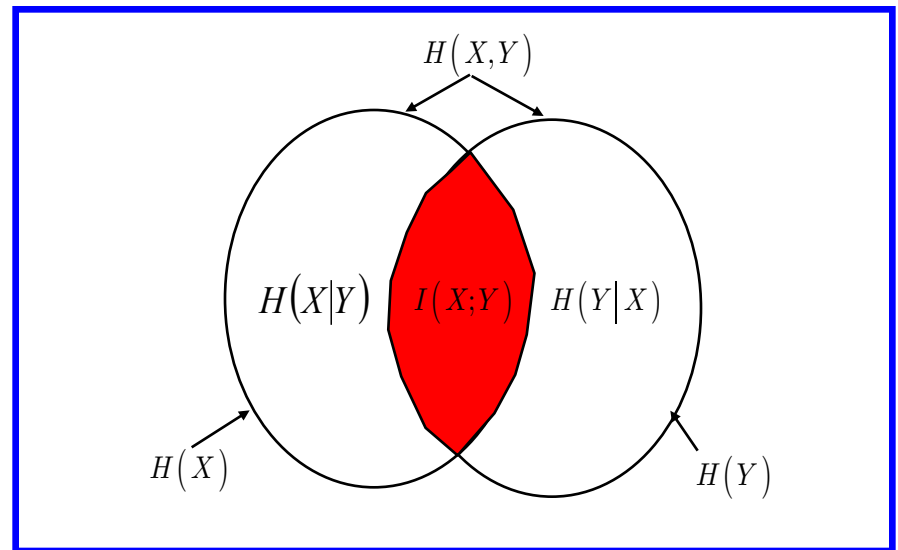
$$I(X;Y) = H(X) - H(X|Y)$$

$$I(X;Y) = H(Y) - H(Y|X)$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

$$I(X;X) = H(X)$$

$$I(X;X) = H(X) - \underbrace{H(X|X)}_0$$



Relationship between mutual information and KLD

$$I(X;Y) = D(p(x,y) \| p(x)p(y)) = E_{p(x,y)} \left[\log \frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)} \right]$$

Mutual Information

Conditional mutual information

$$I(X;Y|Z) = H(X|Z) - H(X|Y,Z)$$

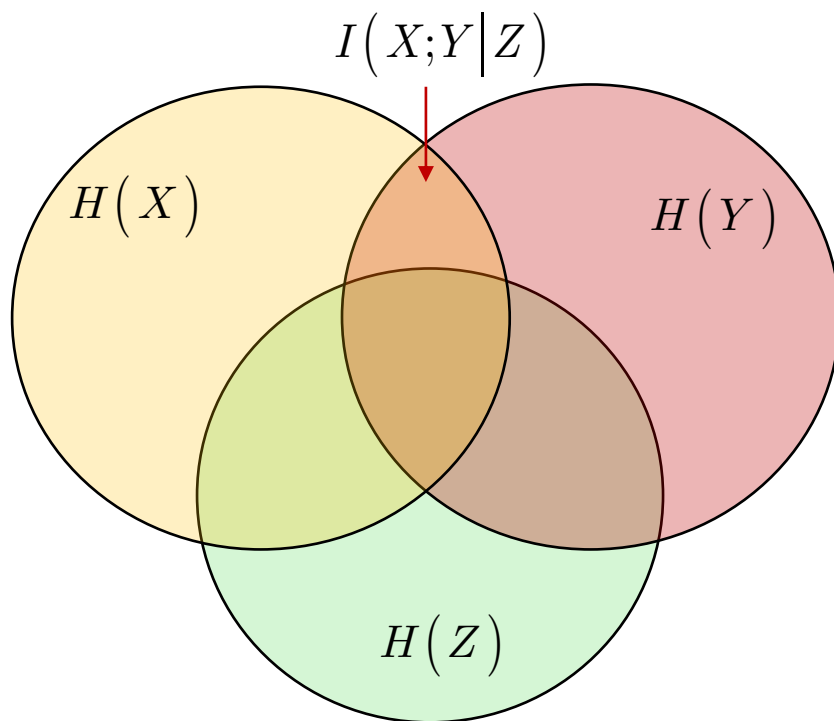
Proof:

$$\begin{aligned} I(X;Y|Z) &= E_{p(x,y,z)} \left[\log \frac{p_{X,Y|Z}(x,y|z)}{p_{X|Z}(x|z)p_{Y|Z}(y|z)} \right] = E_{p(x,y,z)} \left[\log \frac{p_{X|Y,Z}(x|y,z)}{p_{X|Z}(x|z)} \right] \\ &\quad p_{X,Y|Z}(x,y|z) = p_{Y|Z}(y|z)p_{X|Y,Z}(x|y,z) \\ &= \underbrace{E_{p(x,y,z)} \left[\log p_{X|Y,Z}(x|y,z) \right]}_{-H(X|Y,Z)} - \underbrace{E_{p(x,y,z)} \left[\log p_{X|Z}(x|z) \right]}_{H(X|Z)} \end{aligned}$$

Summary: properties of mutual information

- **Conditional mutual information**

$$I(X;Y|Z) = H(X|Z) - H(X|Y,Z)$$



Summary: properties of mutual information

- Chain rule for mutual information

$$I(X_1, X_2, \dots, X_N; Y) = \sum_{i=1}^N I(X_i; Y | X_{i-1}, \dots, X_1)$$

$$I(X_1, X_2, \dots, X_N; Y) = H(X_1, X_2, \dots, X_N) - H(X_1, X_2, \dots, X_N | Y)$$

$$\left\{ I(Z; Y) = H(Z) - H(Z | Y) \right\}$$

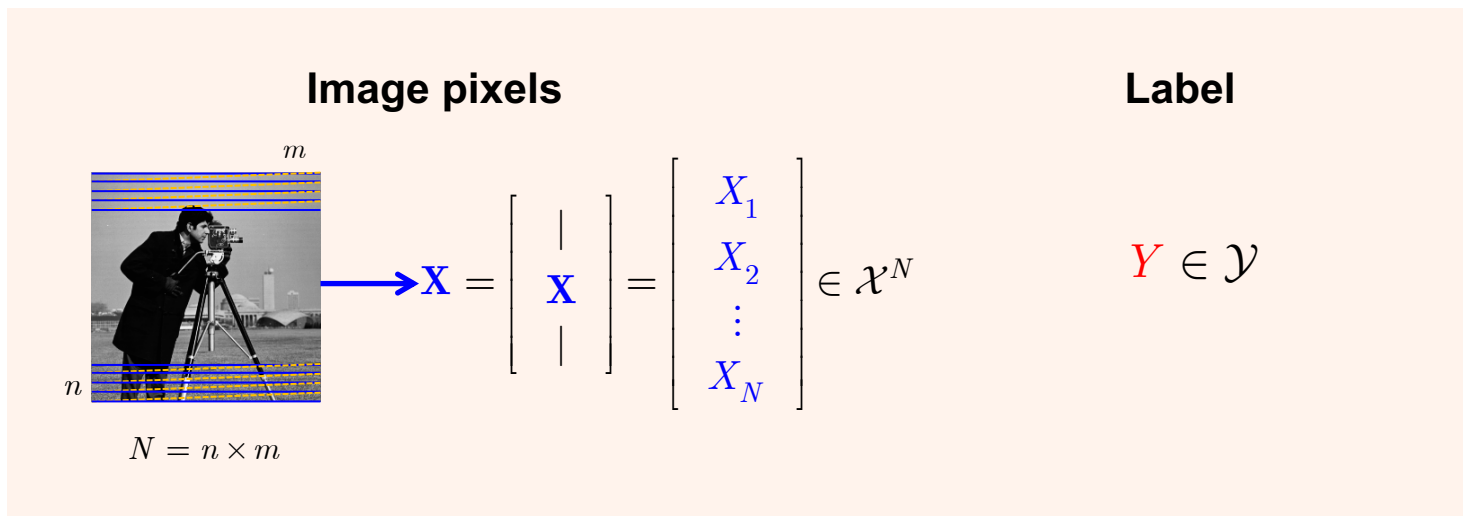
$$\left\{ \begin{array}{l} \text{Chain rule} \\ H(X_1, X_2, \dots, X_N) = \sum_{i=1}^N H(X_i | X_{i-1}, \dots, X_1) \end{array} \right\}$$

$$\begin{aligned} &= \sum_{i=1}^N H(X_i | X_{i-1}, \dots, X_1) - \sum_{i=1}^N H(X_i | X_{i-1}, \dots, X_1, Y) = \\ &= \sum_{i=1}^N I(X_i; Y | X_{i-1}, \dots, X_1) \end{aligned}$$

Summary: properties of mutual information

- Chain rule for mutual information

$$I(X_1, X_2, \dots, X_N; Y) = \sum_{i=1}^N I(X_i; Y | X_{i-1}, \dots, X_1)$$



Mutual information: properties

- Example: more complex decomposition

$$I(X_1, X_2; Y_1, Y_2) = I(X_1; Y_1, Y_2) + I(X_2; Y_1, Y_2 | X_1)$$

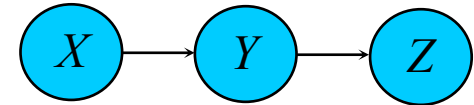
$$I(X_1; Y_1, Y_2) = I(X_1; Y_1) + I(X_1; Y_2 | Y_1) = I(X_1; Y_2) + I(X_1; Y_1 | Y_2)$$

$$\begin{aligned} I(X_2; Y_1, Y_2 | X_1) &= I(X_2; Y_1 | X_1) + I(X_2; Y_2 | X_1, Y_1) \\ &= I(X_2; Y_2 | X_1) + I(X_2; Y_1 | X_1, Y_2) \end{aligned}$$

Summary: properties of mutual information

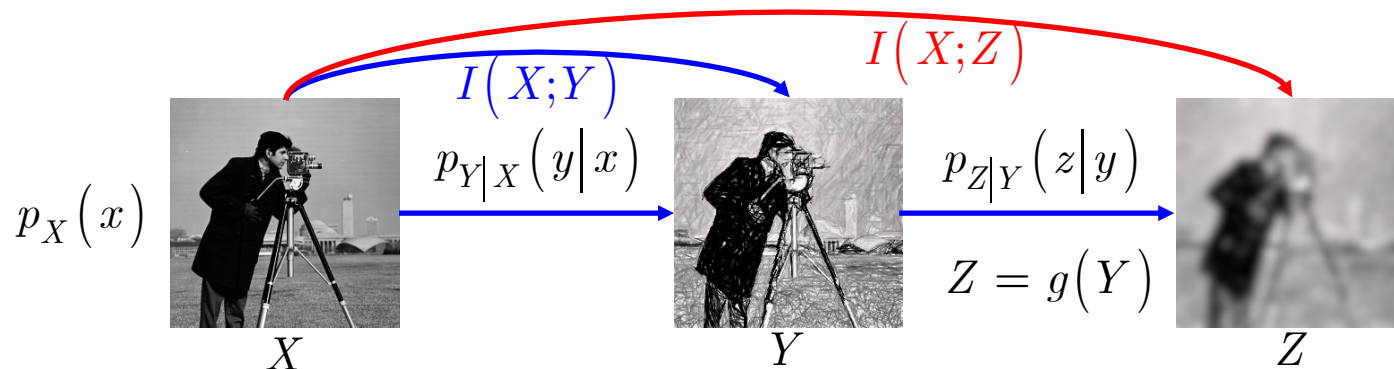
- Non-creativity of information by processing /data processing inequality/

- Consider the Markov chain $p_{X,Y,Z}(x,y,z) = p_X(x)p_{Y|X}(y|x)p_{Z|Y}(z|y)$



- Two inequalities are valid

$$I(X;Y) \geq I(X;Z) \quad I(Y;Z) \geq I(X;Z)$$



Summary: properties of mutual information

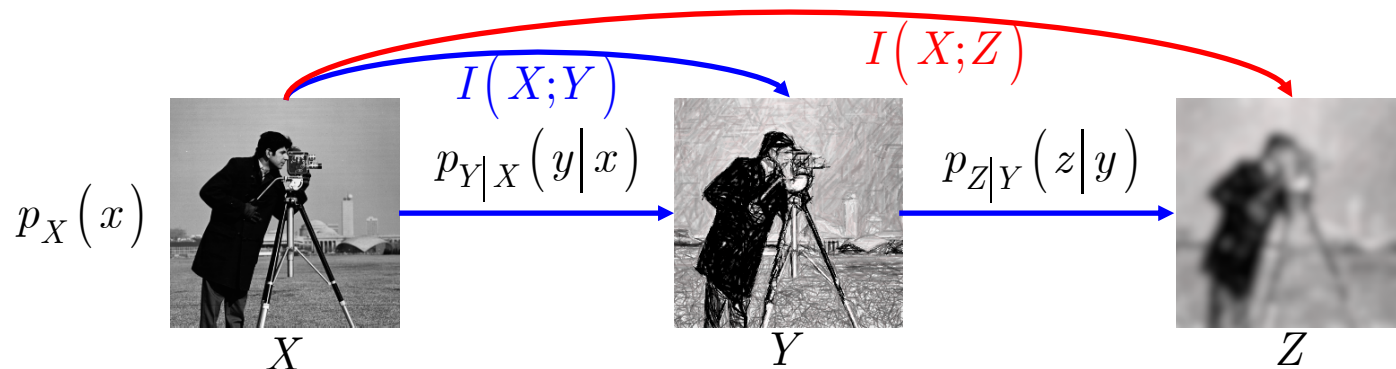
- Data processing inequality

The amount of information cannot be increased by any processing $g(Y)$

This also includes deep networks!

- Two inequalities are valid

$$I(X;Y) \geq I(X;Z) \quad I(Y;Z) \geq I(X;Z)$$



Summary: properties of mutual information

- Proof of data processing inequality
- Using chain rule for mutual information

$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z) \quad (a)$$

$$= I(X;Y) + I(X;Z|Y) \quad (b)$$

$$I(X;Z|Y) = 0 \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \end{array} \begin{array}{c} \text{---} \end{array} \quad X \perp Z|Y$$

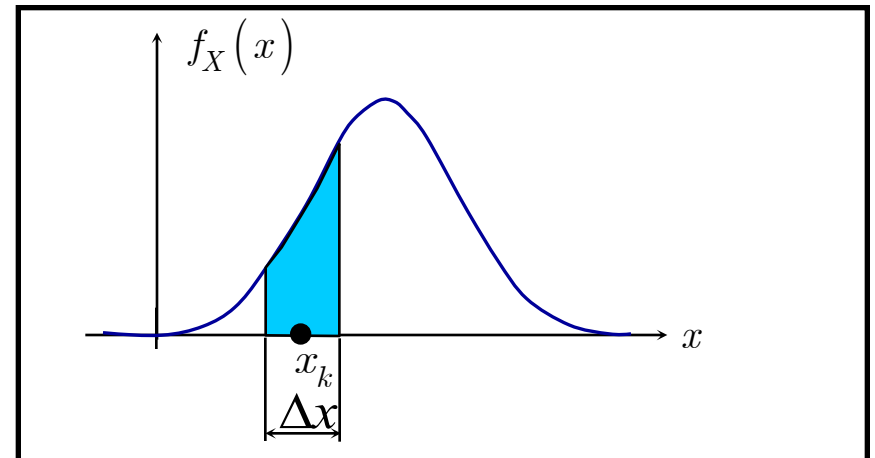
$$I(X;Y) = I(X;Z) + I(X;Y|Z) \Rightarrow I(X;Y) \geq I(X;Z)$$

Continuous R.Vs.: Differential entropy

Definition (entropy): *Differential entropy* of r.v. X with pdf $f_X(x)$

$$h(X) = - \int_{\mathcal{X}} f_X(x) \log_2 f_X(x) dx = E_{f_X} [-\log_2 f_X(x)]$$

$$P[x_k; \Delta x] = \int_{\Delta x} f_X(x) dx \cong f_X(x_k) \Delta x$$



Continuous R.Vs.: Differential entropy

$$\begin{aligned} H(X; \Delta x) &= -\sum_{i=1}^N P[x_k, \Delta x] \log_2 P[x_k, \Delta x] = -\sum_{i=1}^N f_X(x_k) \Delta x \log_2 f_X(x_k) \Delta x \\ &= -\sum_{i=1}^N f_X(x_k) \Delta x \log_2 f_X(x_k) - \underbrace{\sum_{i=1}^N f_X(x_k) \Delta x}_{=1} \underbrace{\log_2 \Delta x}_{const} \\ &\quad \int_{-\infty}^{+\infty} f_X(x) dx = 1 \end{aligned}$$

In the limit Δx tends to zero for large N . As a result, $\log_2 \Delta x$ tends to infinity.

$$H(X; \Delta x) = -\underbrace{\int f_X(x) \log_2 f_X(x) dx}_{h(X)} - \log_2 \Delta x.$$

Continuous R.Vs.: Differential entropy

Definition (Differential entropy of Gaussian r.v.): *differential entropy of Gaussian r.v.* $X \sim \mathcal{N}(0, \sigma_X^2)$ is:

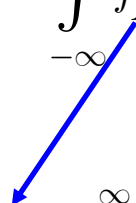
$$h(X) = -E_{f_X} [\log_2 f_X(x)] = \frac{1}{2} \log_2 (2\pi e \sigma_X^2).$$

Proof:

$$\begin{aligned} h(X) &= -\int_{\mathcal{X}} f_X(x) \ln f_X(x) dx \quad [nats] = -\int_{-\infty}^{\infty} f_X(x) \left[-\frac{x^2}{2\sigma_X^2} - \ln \sqrt{2\pi\sigma_X^2} \right] dx = \\ &= \int_{-\infty}^{\infty} f_X(x) \frac{x^2}{2\sigma_X^2} dx + \ln \sqrt{2\pi\sigma_X^2} \int_{-\infty}^{\infty} f_X(x) dx \end{aligned}$$

Continuous R.V.s.: Differential entropy

$$h(X) = \int_{-\infty}^{\infty} f_X(x) \frac{x^2}{2\sigma_X^2} dx + \ln \sqrt{2\pi\sigma_X^2} \underbrace{\int_{-\infty}^{\infty} f_X(x) dx}_{=1}$$


$$\text{Var}[X] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \sigma_X^2$$


$$\int_{-\infty}^{\infty} f_X(x) \frac{x^2}{2\sigma_X^2} dx = \frac{\sigma_X^2}{2\sigma_X^2} = \frac{1}{2}$$

$$h(X) = \frac{1}{2} + \ln \sqrt{2\pi\sigma_X^2} = \frac{1}{2} \ln e + \frac{1}{2} \ln 2\pi\sigma_X^2 = \frac{1}{2} [\ln e + \ln 2\pi\sigma_X^2] = \frac{1}{2} \ln 2\pi e \sigma_X^2 \text{ [nats]}$$

$$h(X) = \frac{1}{2} \log_2 2\pi e \sigma_X^2 \text{ [bits]}$$

Differential entropy: properties

1. Translation does not change the entropy:

$$h(X + a) = h(X).$$

2. Impact of scaling on the differential entropy, if X is scalar r.v.:

$$h(Xa) = h(X) + \log|a|,$$

and if X is a random:

$$h(\mathbf{A}\mathbf{X}) = h(\mathbf{X}) + \log|\det(\mathbf{A})|,$$

determinant of
 $\mathbf{A}$

